

COMPONENTS

Hall sensors

TDK Corporation has expanded its Micronas 3D HAL sensor portfolio with HAR 3900 and HAR 3930. The products enable stray-field compensated position detection in automotive and industrial



applications. They feature 3D magnetic-field measurement capability and 2D stray-field

robust position detection. Both sensors are the dual-die SMD package versions of HAL 3900 and HAL 3930, and suit a wide range of applications including steering-angle position detection, transmission position detection, shifter position detection, accelerator and brake-pedal position detection. The HAR 39xy sensors enable angular measurements up to 360° with ferrite 2-pole magnets in addition to linear measurements up to 35mm by using two-pole bar magnets.

TDK Corporation

www.micronas.tdk.com

MCUs for contactless operation

Renesas Electronics has introduced the RX671 group of 32-bit microcontrollers (MCUs) that provide high-performance



touch sensing and voice recognition capabilities for contactless operation.

The RX671 are built around an RXv3 CPU core operating at 120MHz and integrate flash memory that support fast read access at a clock speed of 60MHz. This gives excellent real-time performance with a CoreMark score of 707 and good power efficiency at 48.8 CoreMark/mA. The MCUs are well suited for a wide range of applications that require advanced functionality,

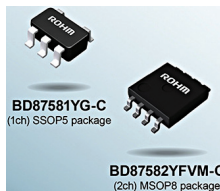
power efficiency, and compact size, such as heating, ventilation, and air-conditioning (HVAC), smart meters, and smart home appliances.

Renesas Electronics Corporation

www.renesas.com

Op-amps with noise immunity

Advancements in electrification and mounting density have increased the amount of environmental noise in electric vehicles and cars equipped with ADAS. It is extremely difficult to carry out noise evaluation on individual boards and systems during automotive development since it must be performed after



assembly. To address this, ROHM has developed the BD8758xY series (BD87581YG-C, BD87582YFVM-C) of rail-to-rail input/output high-speed CMOS op-amps that feature improved EMI immunity for automotive and industrial equipment applications requiring high-speed sensing in harsh environments. Due to their superior noise immunity, the design prevents malfunctions and reduces design resources by combining a vertically integrated production system with proprietary analogue design technology.

ROHM Semiconductor

www.rohm.com

BLDC motor drivers

Texas Instruments (TI) has introduced 70W brushless DC (BLDC) motor drivers to provide code-free, sensorless trapezoidal and field-oriented control. The devices enable engineers to spin a BLDC motor in less than 10 minutes—eliminating months of design time for industrial systems such as major and small home

appliances, and medical applications such as ventilators and continuous positive airway pressure machines.



With integrated real-time control and up to 18 discrete components

including power MOSFETs, the devices also speed system response time while reducing board space by as much as 70%, all while providing best-in-class acoustic performance.

Texas Instruments

www.ti.com

RF LDMOS power transistor

STMicroelectronics is adding a broad range of new devices to the STPOWER family of LDMOS transistors, which comprises three different product series



optimised for RF power amplifiers in a variety of commercial and industrial applications. Featuring

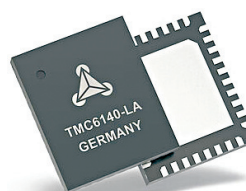
high efficiency and low thermal resistance, and packaged to handle high RF power, the transistors combine a short conduction-channel length with a high breakdown voltage. These characteristics permit a cost-effective solution with low power consumption and high reliability.

STMicroelectronics

www.st.com

3-phase MOSFET gate driver

TRINAMIC Motion Control introduces the fully integrated TMC6140-LA 3-phase MOSFET gate driver that simplifies design and maximises battery life



of brushless direct current (DC) motor drives. The gate driver integrates all three bottom